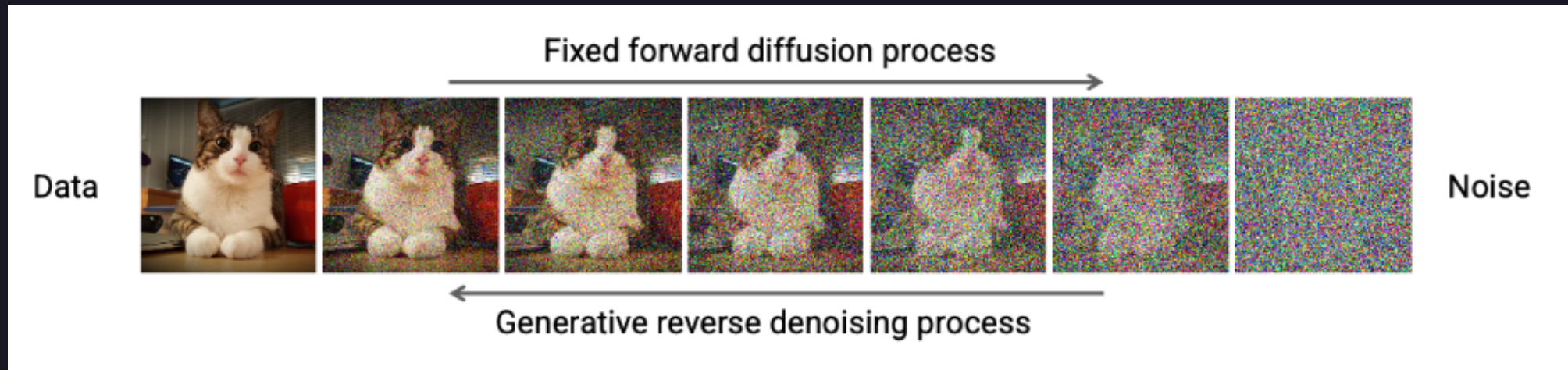


Boosting Generative Image Modeling via Joint Image Feature Synthesis

Theodoros Kouzelis, Eystathis Karypidis, Ioannis Kakogeorgiou,
Spyros Gidaris, Nikos Komodakis

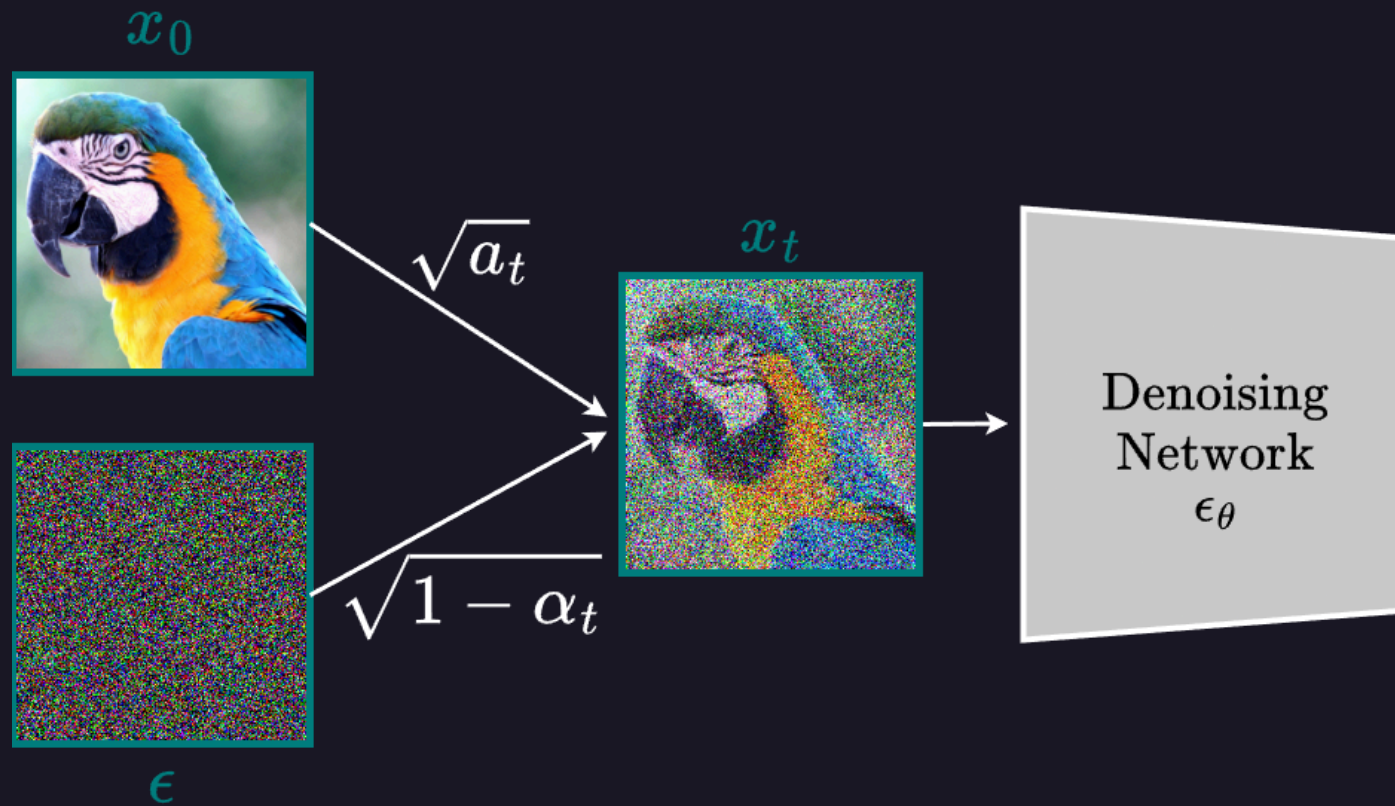
Diffusion / Flow Models (in a nutshell)

- Given an Image $\mathbf{x}$ and Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$
- Construct a time dependent noising process:
 - $\mathbf{x}_t = \alpha_t \mathbf{x} + \sigma_t \epsilon$
- Reverse process to generate data from noise:
 - $\mathbf{x}_{t-1} = \tilde{\mathbf{x}}_t + \mathbf{w}_t \cdot \text{Network Output}$



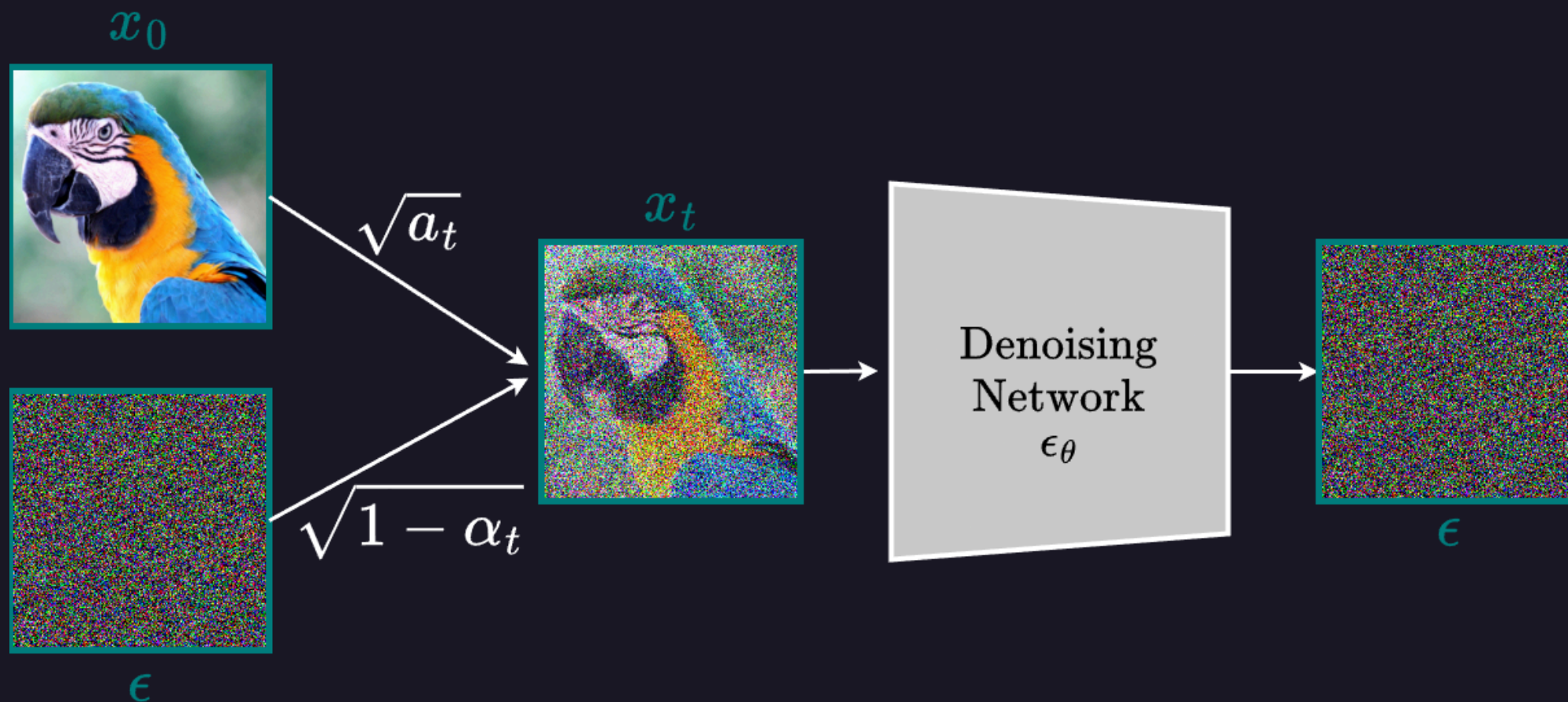
Denoising Objective (DDPM)

- Construct training sample $\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x} + \sqrt{1 - \alpha_t}\boldsymbol{\epsilon}$



Denoising Objective (DDPM)

- Loss function $\mathcal{L}_{\text{simple}} = \mathbb{E}_x \|\epsilon_{\theta}(\mathbf{x}_t) - \epsilon\|_2^2$

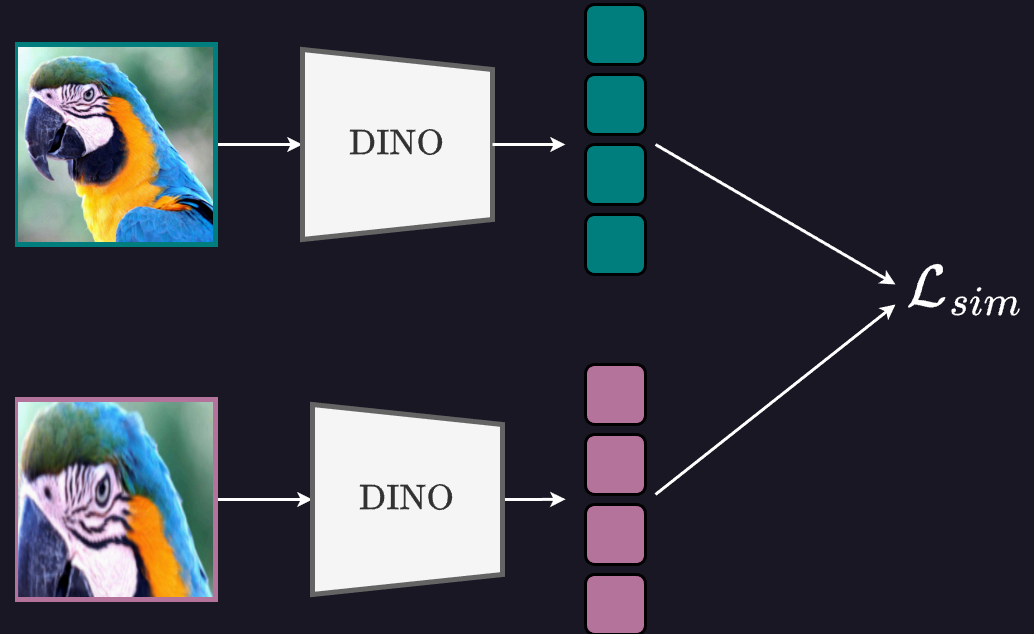


The problem with the Denoising Objective

- The goal is purely to remove noise, regardless of the content.
- Doesn't distinguish between meaningful semantics and irrelevant image details.
- Does not result in (very) good representations

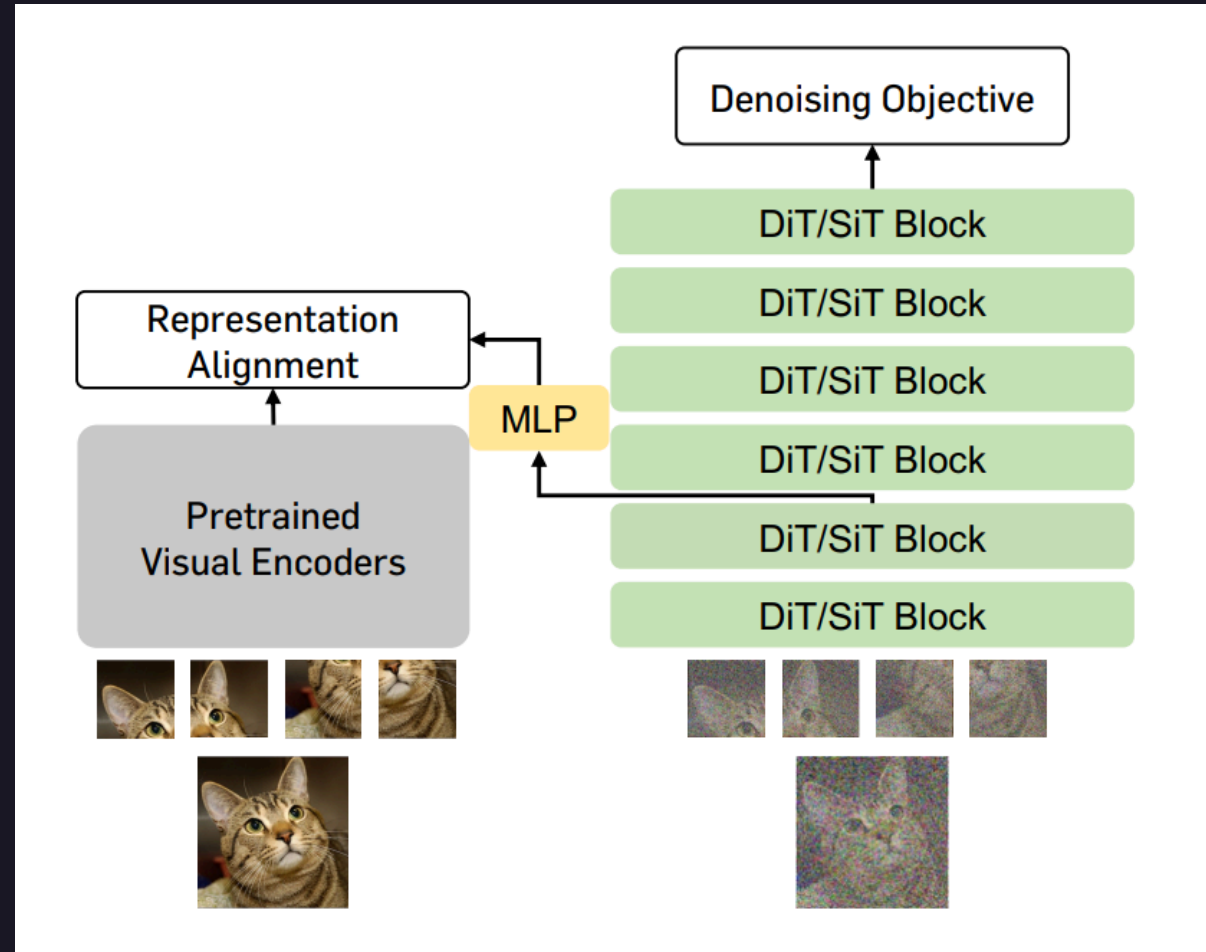
DINO (in a nutshell)

- Take different "views" from an image
- Force representations to be similar
- Very strong representations



Representation Alignment

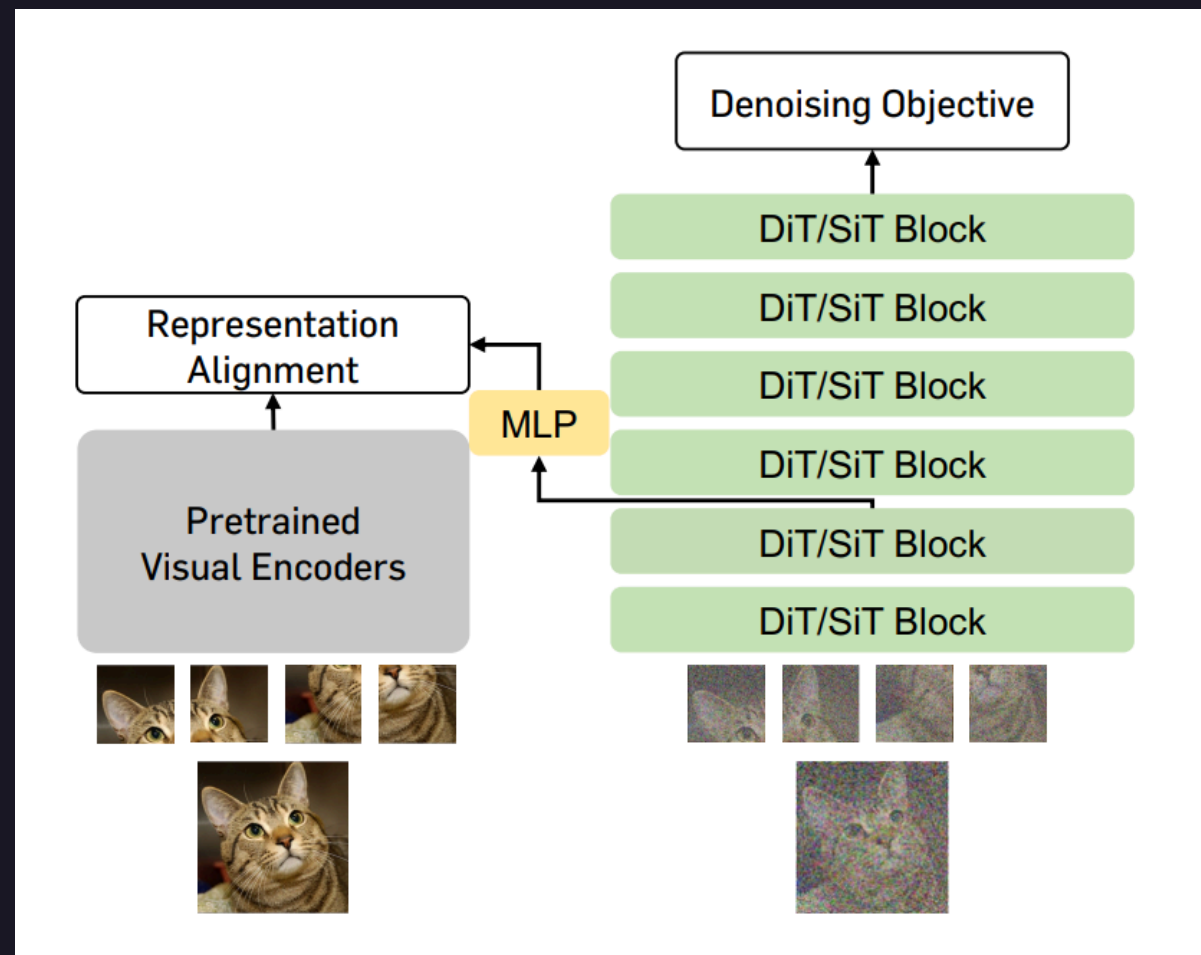
- Align features with the representations of powerful visual encoders.



Representation Alignment

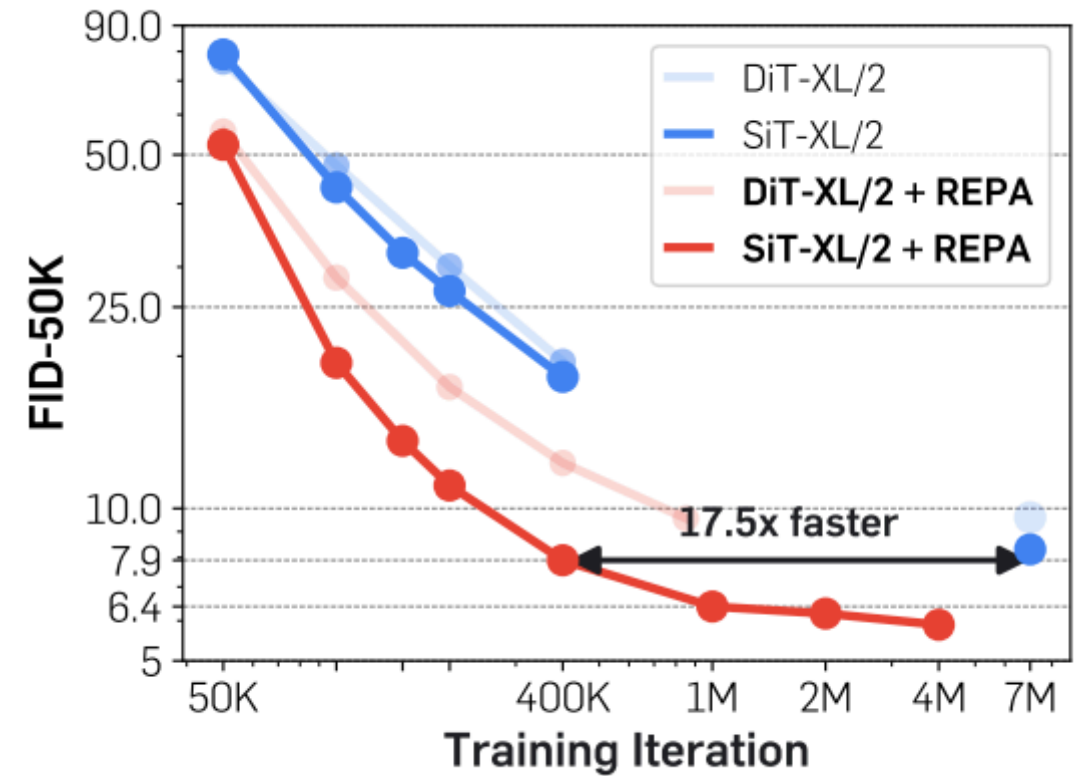
- Let h_t be the an intermediate feature
- $y_* = f(x)$ (e.g. DINOv2)
- and h_ϕ a simple projection layer
- REPA loss:

- $-\mathbb{E}[\frac{1}{N} \sum_i \text{sim}((y_*^i), h_\phi(h_t^i))]$



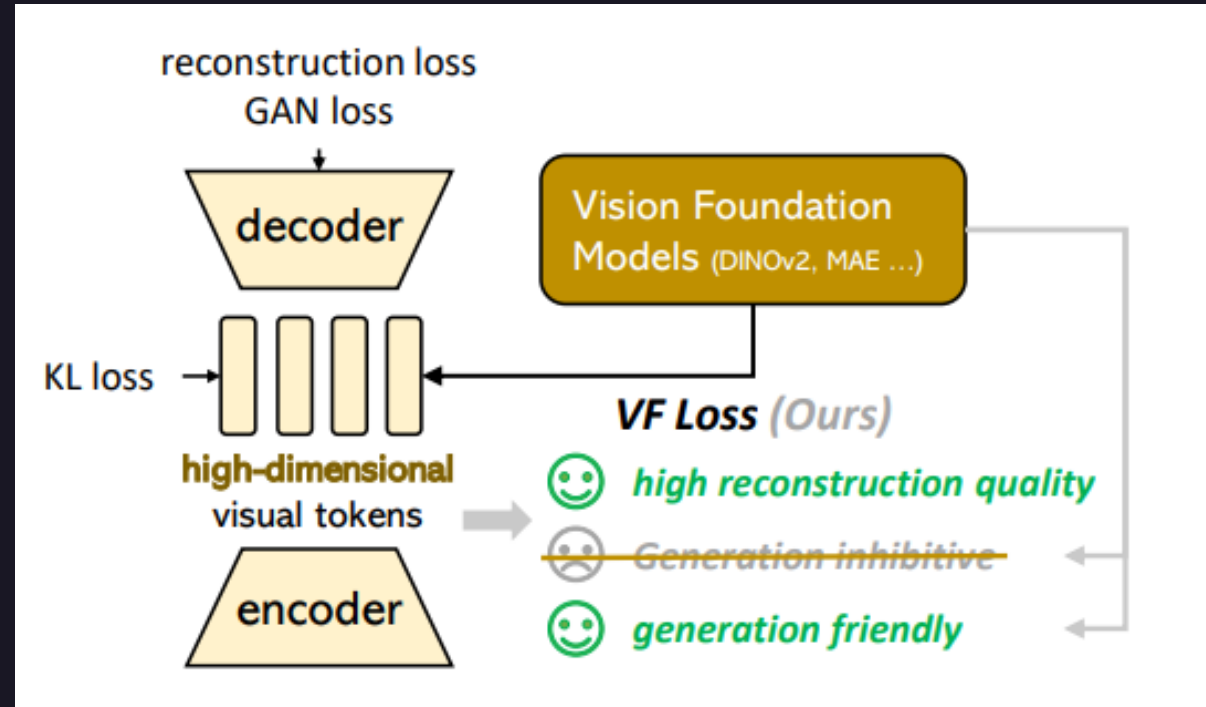
Representation Alignment

- $\times 17$ faster convergence
- Better generative performance

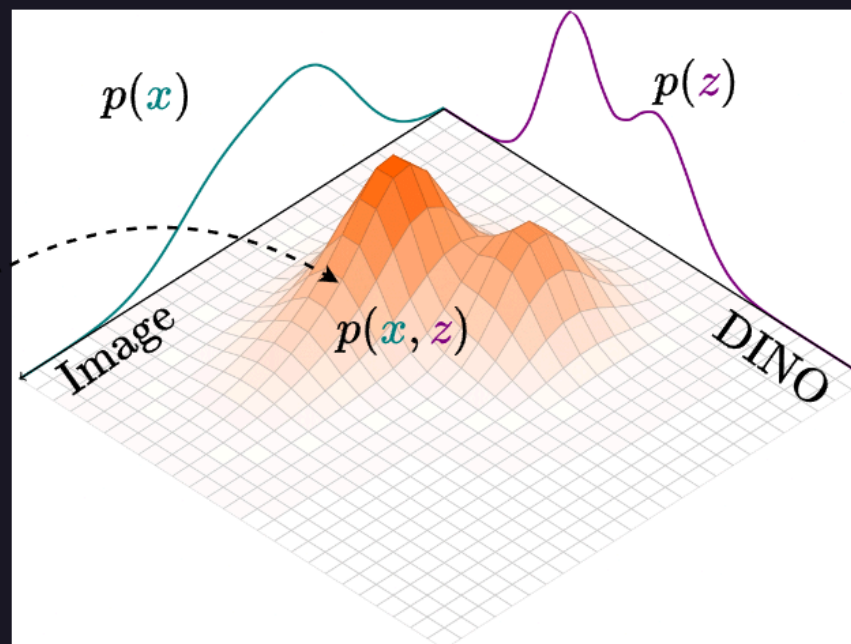
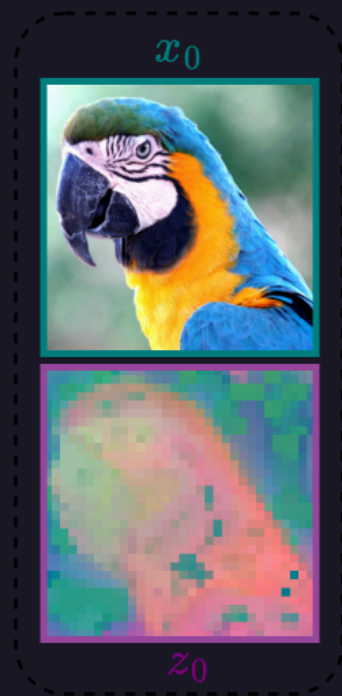


Representation Alignment for VAE Latents

- Aligning the **VAE latents** with DINOv2 also results in better generative performance

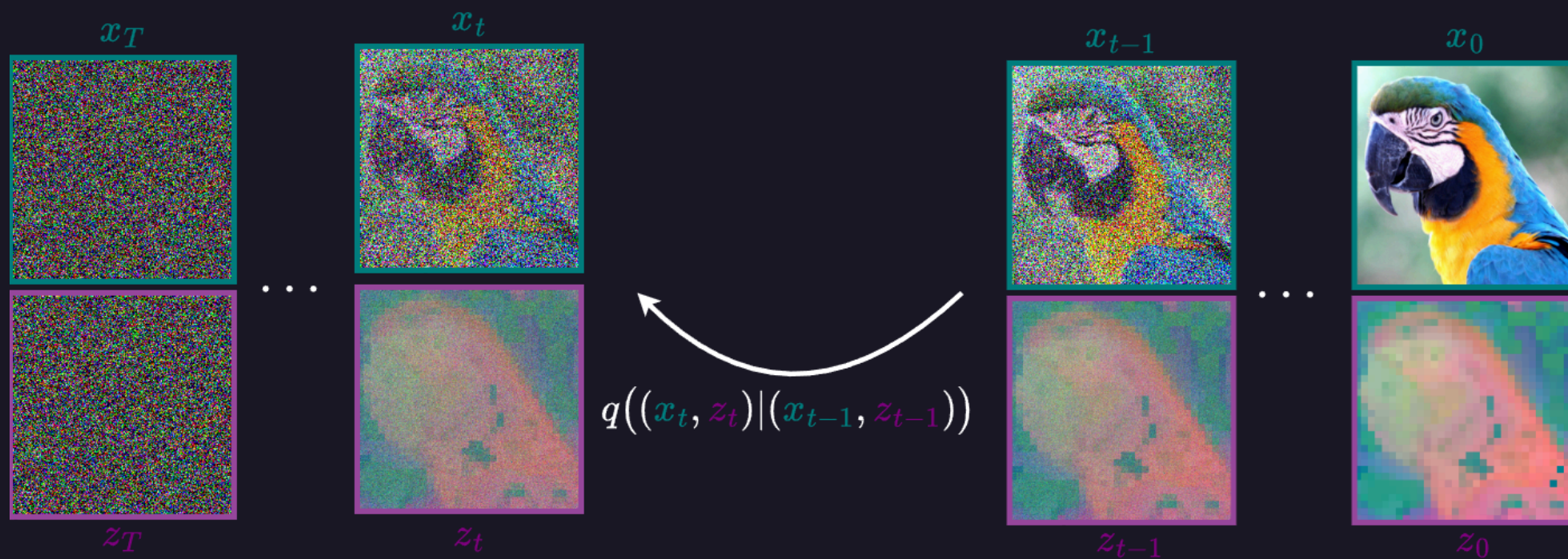


ReDi: Joint Image-Feature Synthesis



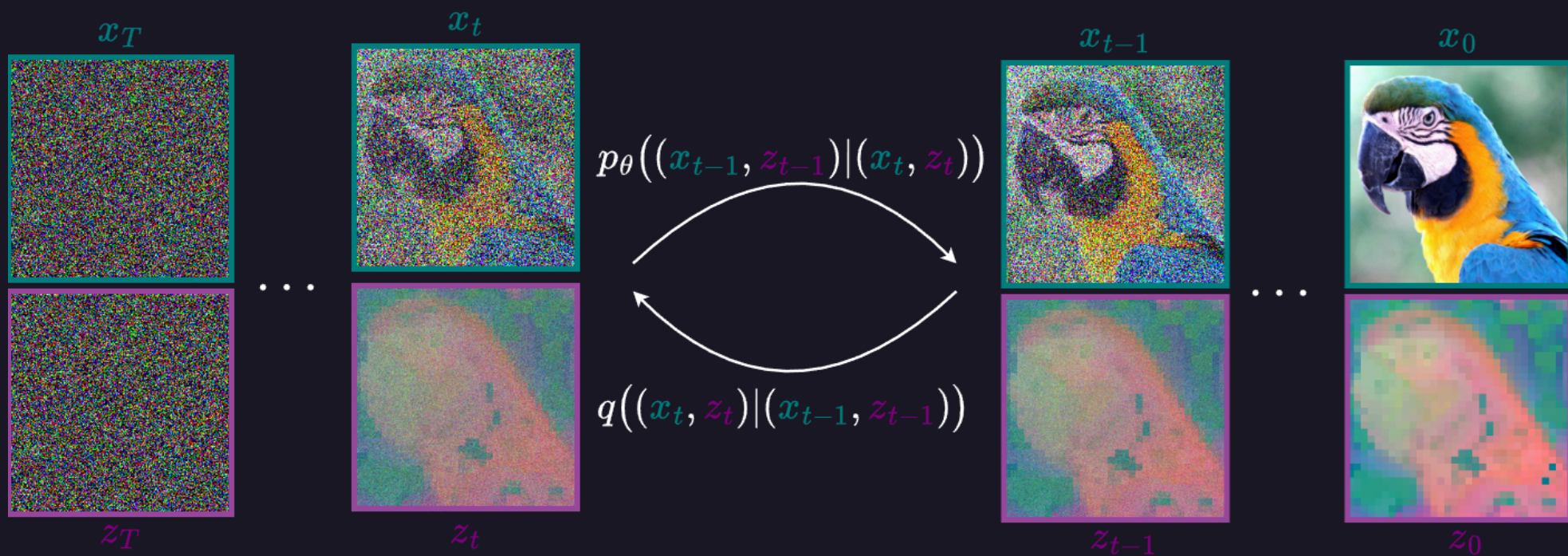
ReDi: Joint Image-Feature Synthesis

Joint Forward Process

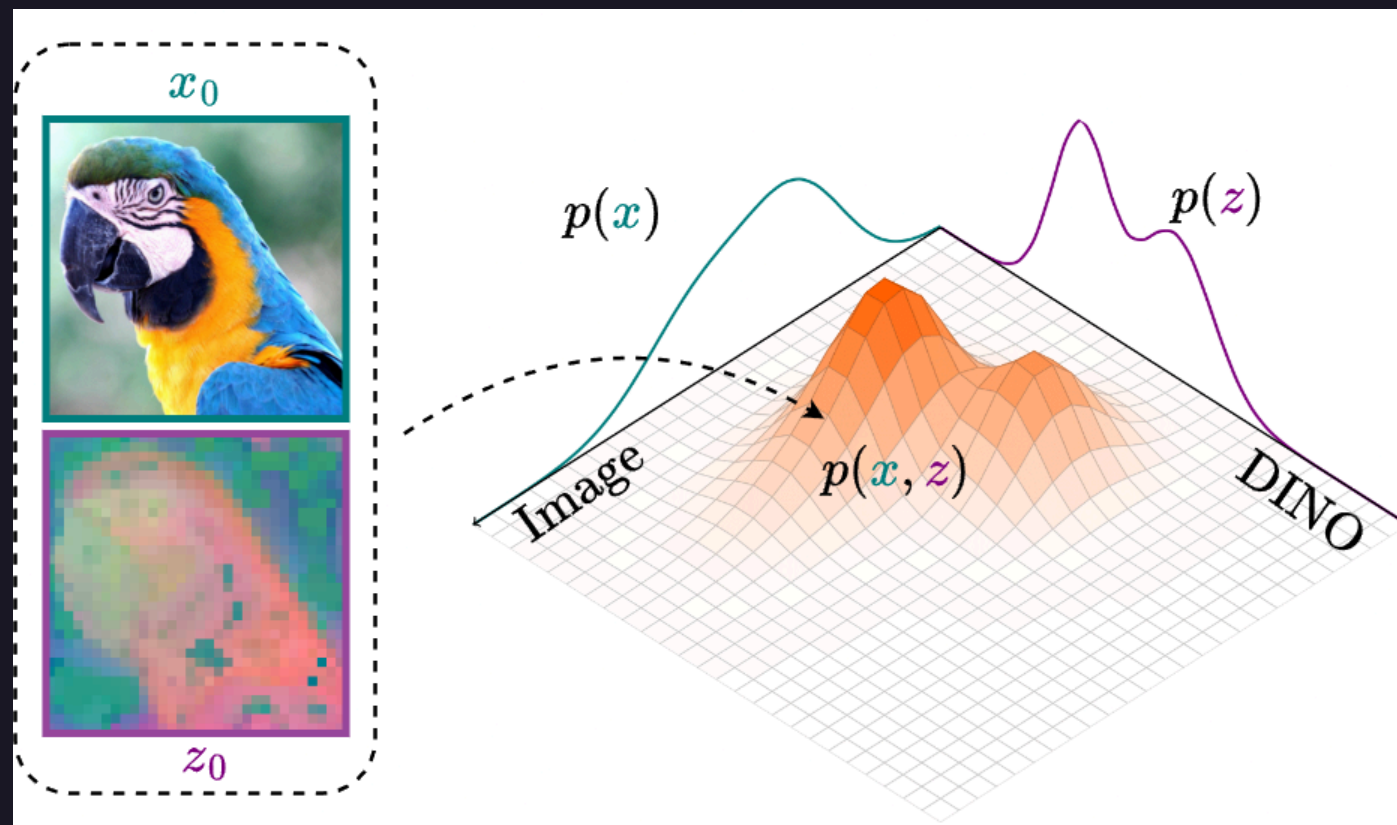


ReDi: Joint Image-Feature Synthesis

Joint Reverse Process

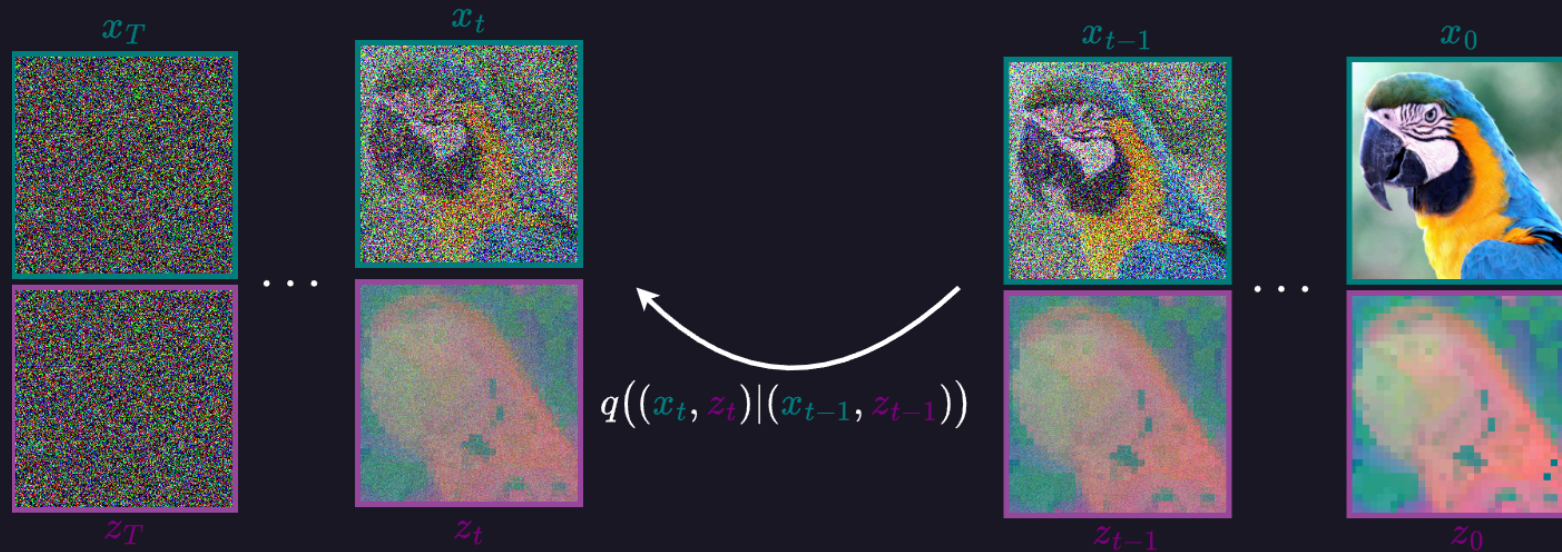


ReDi: Joint Image-Feature Synthesis



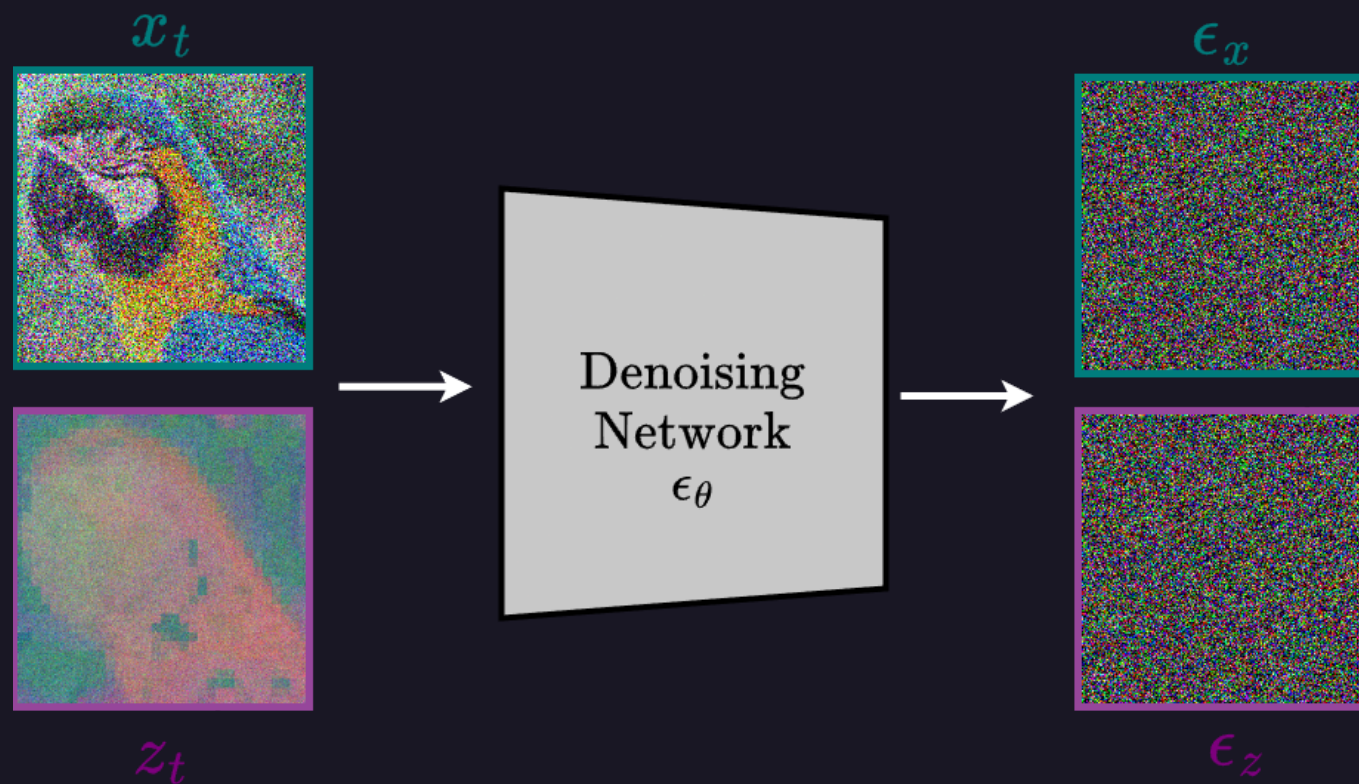
Joint Forward Process

- VAE Latents: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_x$
- DINOv2: $\mathbf{z}_t = \sqrt{\bar{\alpha}_t} \mathbf{z}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_z$



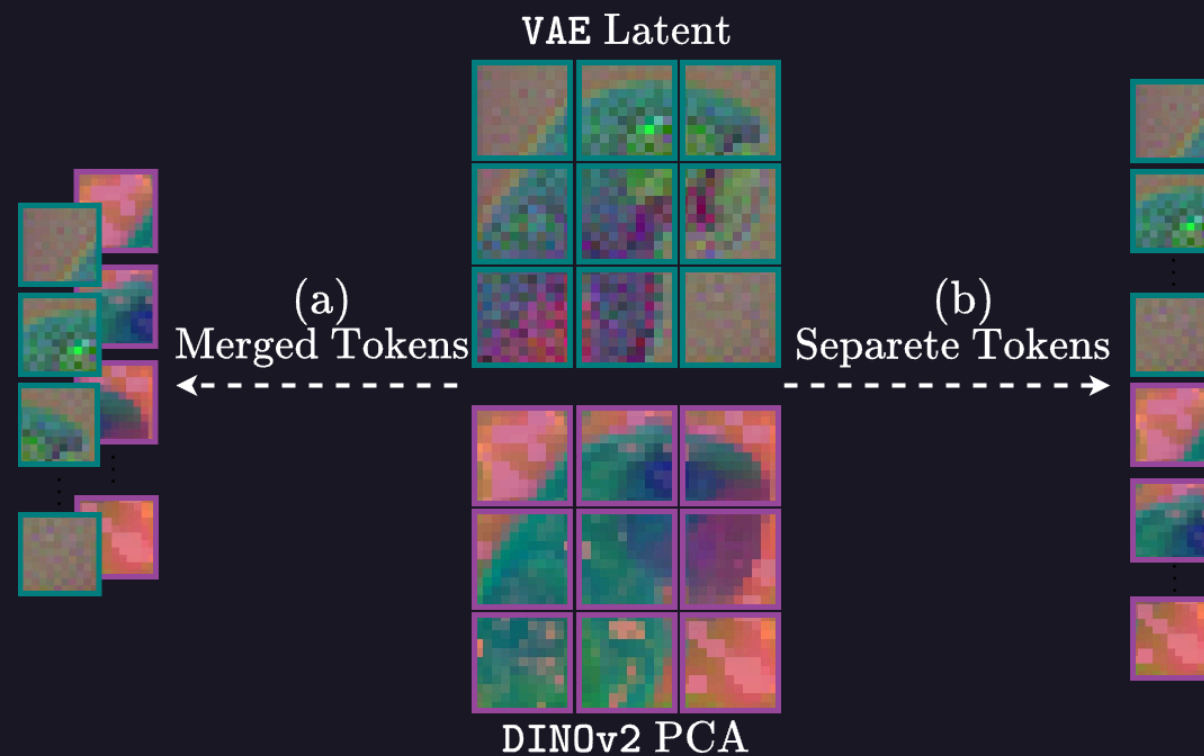
Joint Denoising Objective

- $$\mathcal{L}_{joint} = \mathbb{E}_{\mathbf{x}_0, \mathbf{z}_0} \left[\|\boldsymbol{\epsilon}_{\theta}^x(\mathbf{x}_t, \mathbf{z}_t, t) - \boldsymbol{\epsilon}_x\|_2^2 + \|\boldsymbol{\epsilon}_{\theta}^z(\mathbf{x}_t, \mathbf{z}_t, t) - \boldsymbol{\epsilon}_z\|_2^2 \right]$$



Fusing the two modelities in a single input sequence

- Merged Tokens
- Separete Tokens



Fusing the two modelities in a single input sequence

- Modality-specific **embedding** matrices:

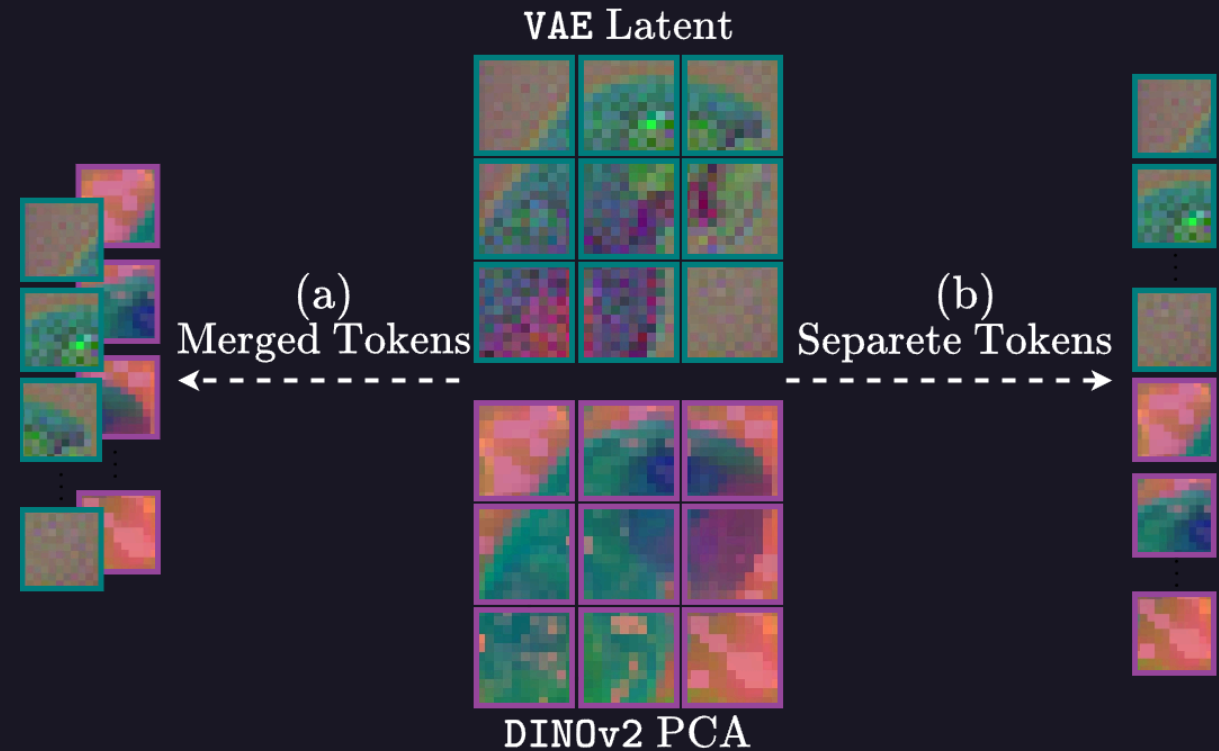
- $\mathbf{W}_{\text{emp}}^x \in \mathbb{R}^{C_x \times C_d}$

- $\mathbf{W}_{\text{emp}}^z \in \mathbb{R}^{C_z \times C_d}$

- Modality-specific **projection** matrices:

- $\mathbf{W}_{\text{dec}}^x \in \mathbb{R}^{C_d \times C_x}$

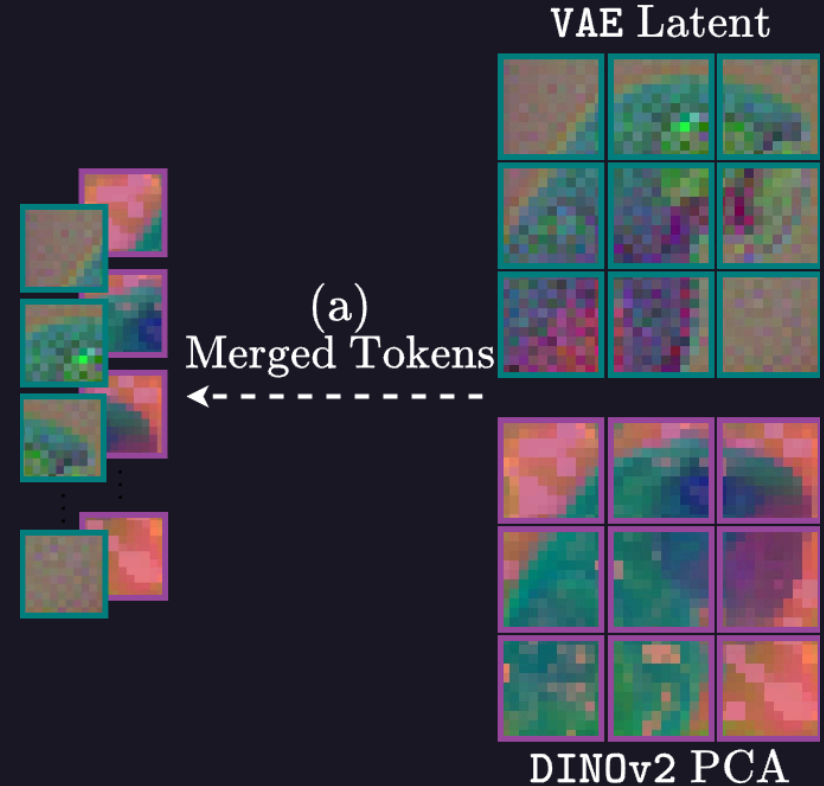
- $\mathbf{W}_{\text{dec}}^z \in \mathbb{R}^{C_d \times C_x}$



Merged Tokens

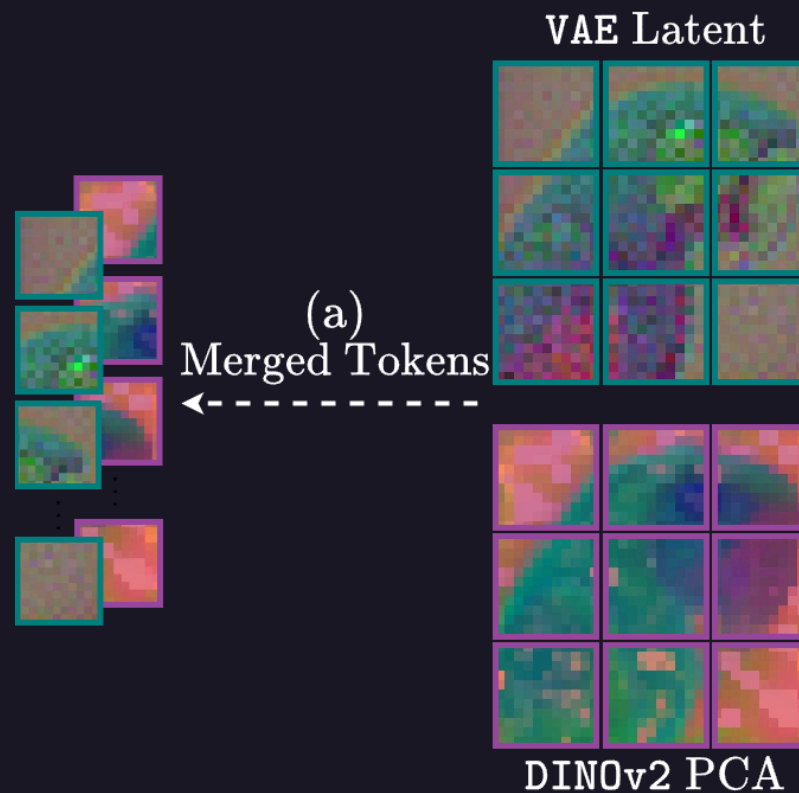
The tokens are summed channel-wise:

- $\mathbf{h}_t = \mathbf{x}_t \mathbf{W}_{\text{emb}}^x + \mathbf{z}_t \mathbf{W}_{\text{emb}}^z \in \mathbb{R}^{L \times C_d}$
- The transformer processes h_t to produce o_t
- $\epsilon_{\theta}^x = \mathbf{o}_t \mathbf{W}_{\text{dec}}^x, \quad \epsilon_{\theta}^z = \mathbf{o}_t \mathbf{W}_{\text{dec}}^z.$



Merged Tokens

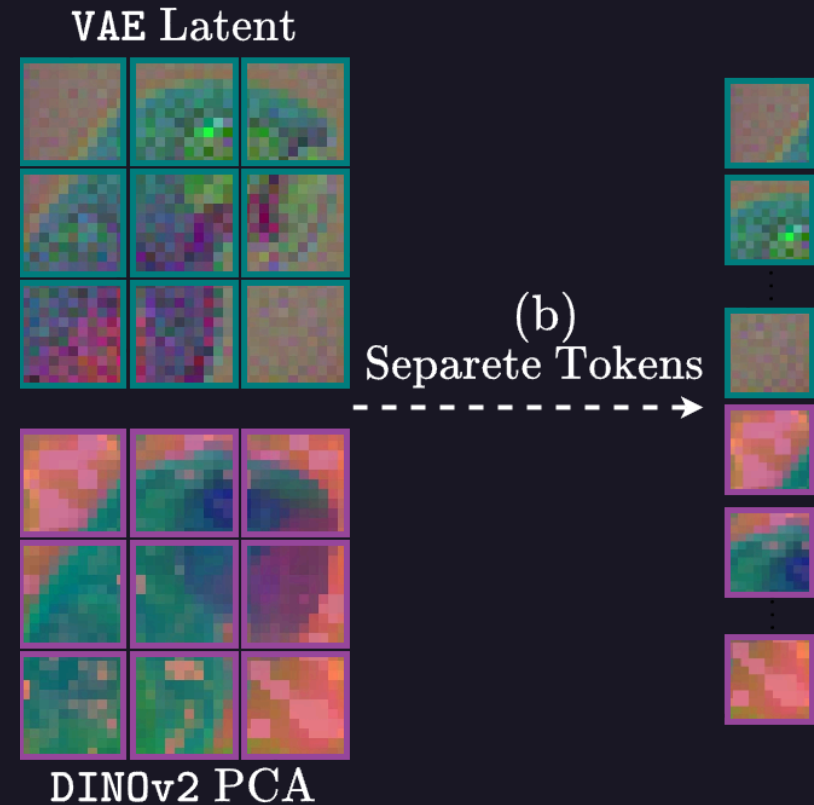
- Early fusion.
- Maintains computational efficiency, as the token count remains unchanged.



Seperate Tokens

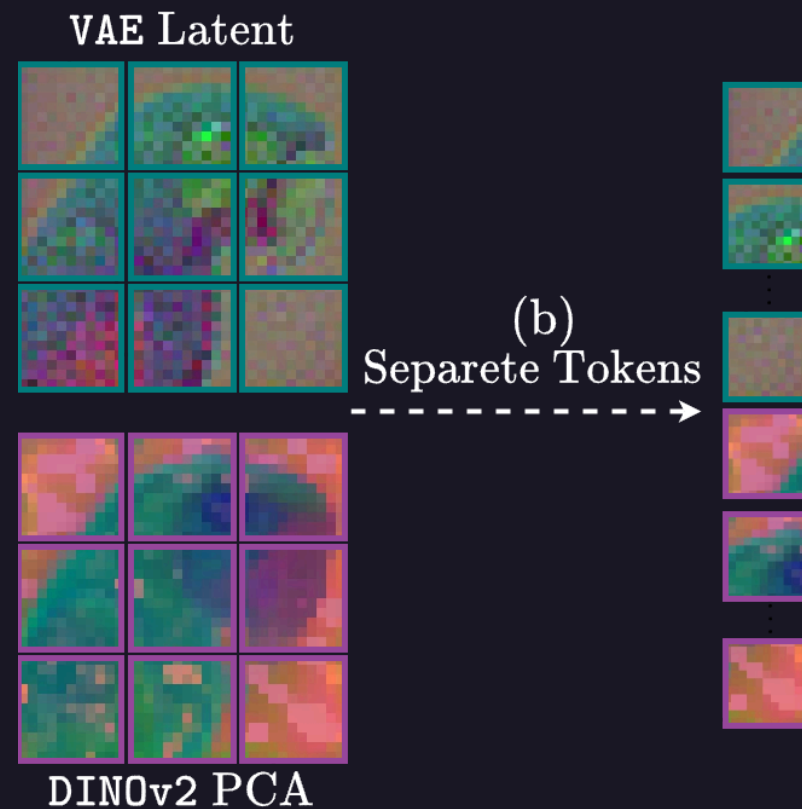
Tokens are concatenated along the sequence dimension:

- $\mathbf{h}_t = [\mathbf{x}_t \mathbf{W}_{\text{emb}}^x, \mathbf{z}_t \mathbf{W}_{\text{emb}}^z] \in \mathbb{R}^{2L \times C_d},$
- The transformer outputs separate representations
 $\mathbf{o}_t = [\mathbf{o}_t^x, \mathbf{o}_t^z]$
- $\boldsymbol{\epsilon}_\theta^x = \mathbf{o}_t^x \mathbf{W}_{\text{dec}}^x, \quad \boldsymbol{\epsilon}_\theta^z = \mathbf{o}_t^z \mathbf{W}_{\text{dec}}^z.$



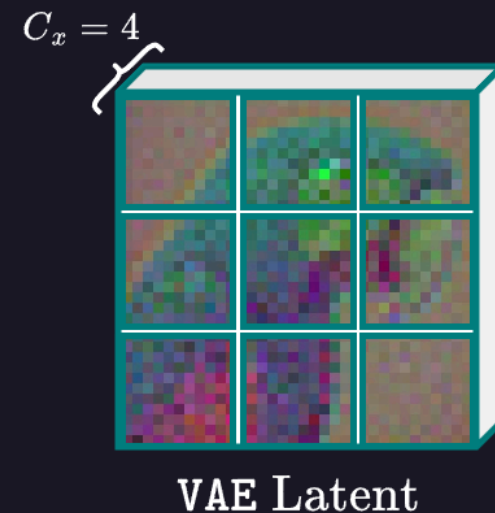
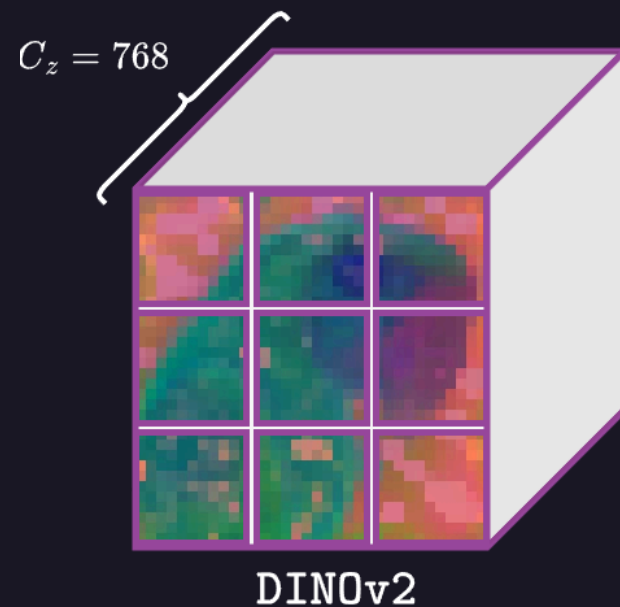
Seperate Tokens

- Preserves modality-specific information.
- **Increased computation** due to increased token count ($\times 2$).



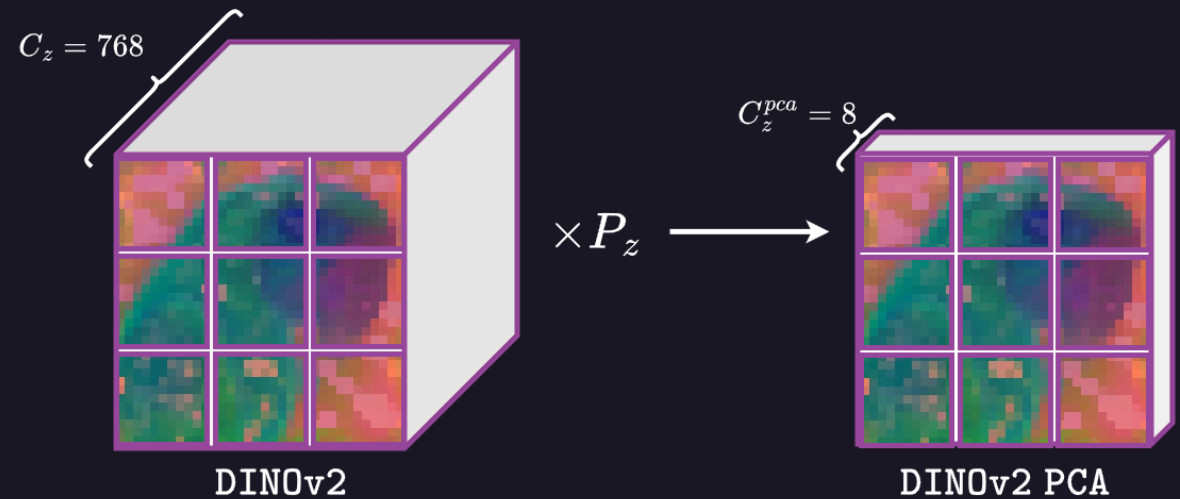
Dimensionality-Reduced Visual Representation

- The channels of DINOv2 significantly exceeds that of VAE latent
- Visual representations dominate model capacity at the expense of image latents.



Dimensionality-Reduced Visual Representation

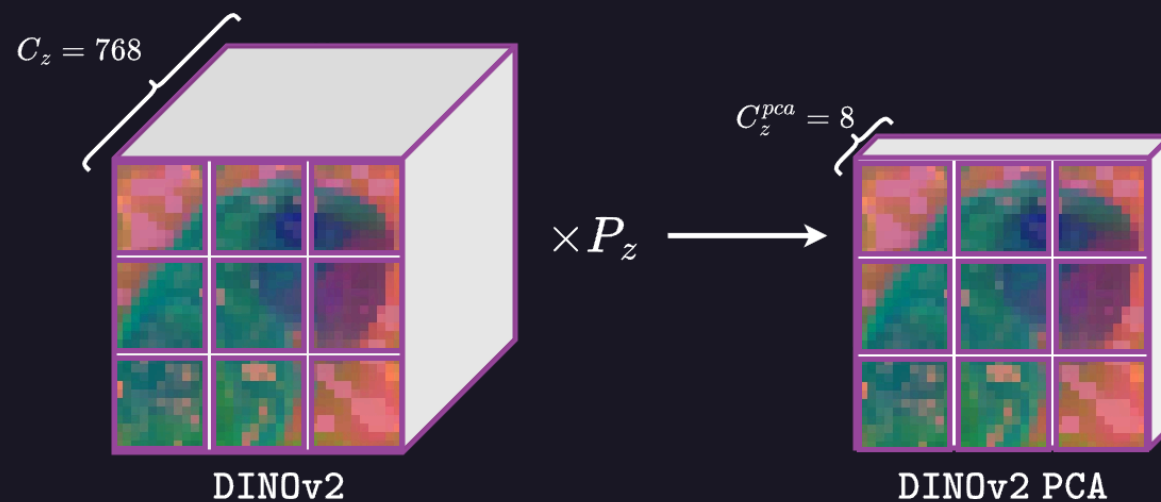
- Sample N features from training set and calculate a PCA projection matrix P_z



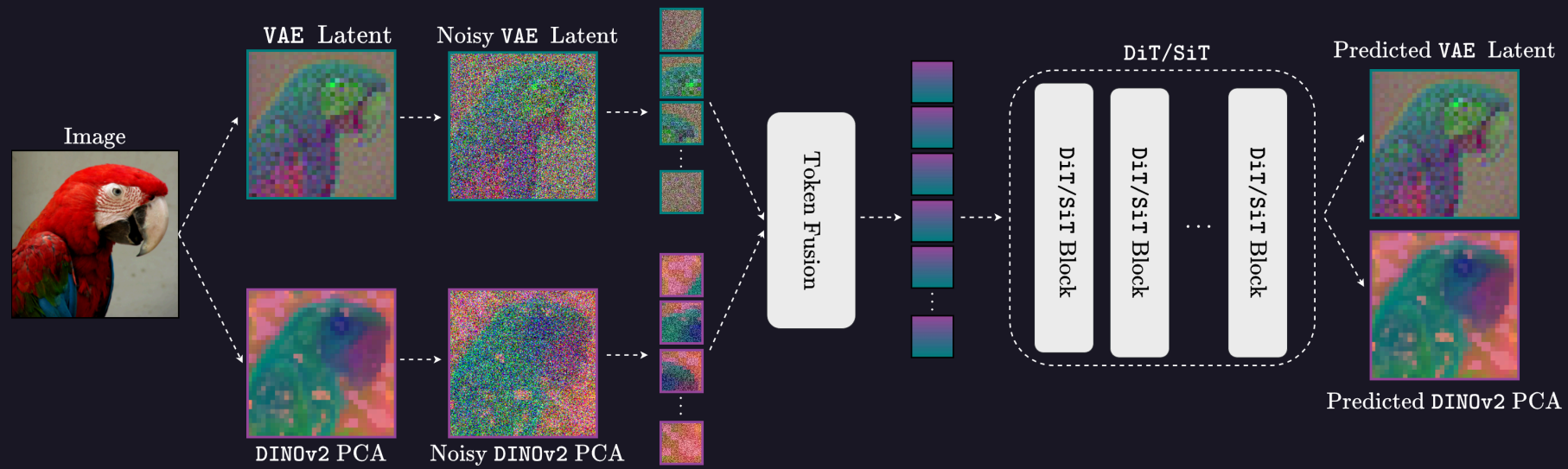
Dimensionality-Reduced Visual Representation

- Project to principal subspace:

$$z_{pca} = z \times P_z$$



Redi Pipeline



Representation Guidance

- Same as Classifier-Free Guidance but with Semantic Representation as the condition.
- During inference we modify the posterior distribution to:
$$\hat{p}_{\theta}(\mathbf{x}_t, \mathbf{z}_t) \propto p_{\theta}(\mathbf{x}_t)p(\mathbf{z}_t|\mathbf{x}_t)^{w_r}$$
- Samples are pushed toward higher likelihood of the conditional distribution
$$p_{\theta}(\mathbf{z}_t|\mathbf{x}_t)$$

Representation Guidance

$$\nabla_{\mathbf{x}_t} \log \hat{p}_\theta(\mathbf{x}_t, \mathbf{z}_t) = \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t) + w_r (\nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t, \mathbf{z}_t) - \nabla_{\mathbf{x}_t} \log p_\theta(\mathbf{x}_t)).$$

- From the equivalence between denoisers and scores we have:

$$\hat{\epsilon}_\theta(\mathbf{x}_t, \mathbf{z}_t, t) = \epsilon_\theta(\mathbf{x}_t, t) + w_r (\epsilon_\theta(\mathbf{x}_t, \mathbf{z}_t, t) - \epsilon_\theta(\mathbf{x}_t, t)).$$

Representation Guidance

$$\hat{\epsilon}_{\theta}(\mathbf{x}_t, \mathbf{z}_t, t) = \epsilon_{\theta}(\mathbf{x}_t, t) + w_r (\epsilon_{\theta}(\mathbf{x}_t, \mathbf{z}_t, t) - \epsilon_{\theta}(\mathbf{x}_t, t)).$$

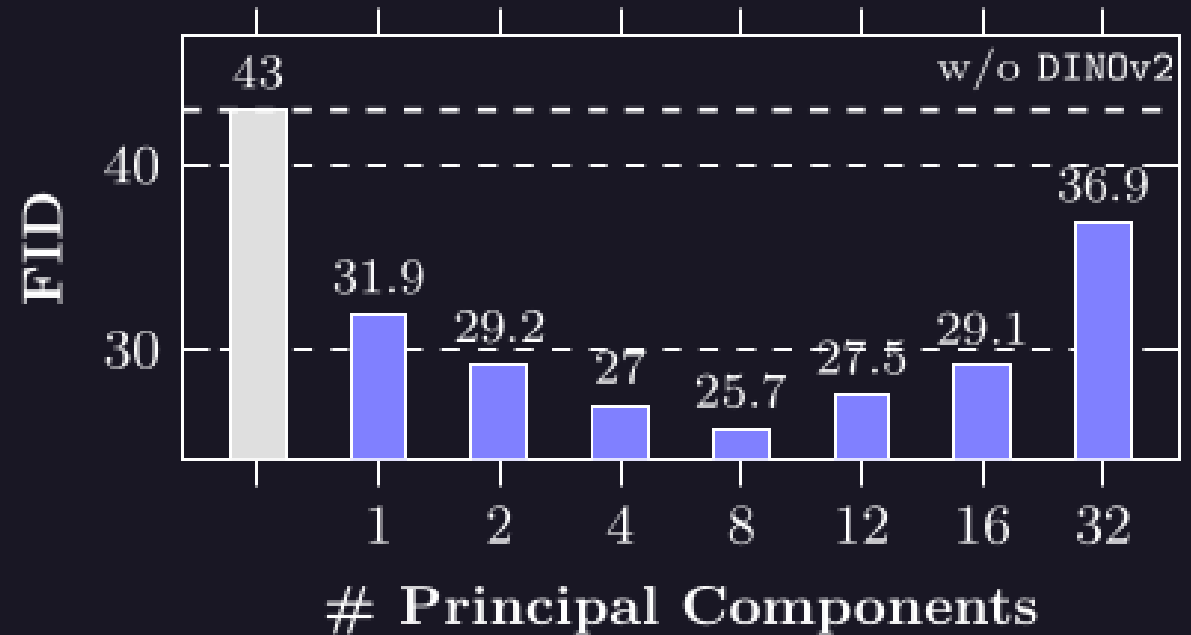
- We train both $e_{\theta}(\mathbf{x}_t, \mathbf{z}_t, t)$ and $e_{\theta}(\mathbf{x}_t, t)$ jointly.
- With probability $p_{drop} = 0.2$:
 - Zero out $\mathbf{z}_t$ (setting $\epsilon_{\theta}(\mathbf{x}_t, t) = \epsilon_{\theta}(\mathbf{x}_t, \mathbf{0}, t)$)
 - Disable the visual representation denoising loss.

Experimental Results

Ablation

Dimensionality reduction

- Intermediate subspace ($r = 8$)
- Retains sufficient expressivity to guide generation
- Does not dominate model capacity.



Ablation

Merged Tokens vs. Separate Tokens.

MODEL	#TOKENS	THROUGHPUT $\uparrow$	FID $\downarrow$
DiT-B/2	256	4.52	43.5
w/ ReDi (MR)	256	4.51	25.7
w/ ReDi (SP)	512	2.26	24.7

Merged Tokens vs. Separate Tokens.

Merged Tokens

- ~Same training and inference compute

MODEL	#TOKENS	THROUGHPUT ↑	FID↓
DiT-B/2	256	4.52	43.5
w/ ReDi (MR)	256	4.51	25.7
w/ ReDi (SP)	512	2.26	24.7

Merged Tokens vs. Separate Tokens.

Merged Tokens

- ~Same training and inference compute

Separate Tokens

- $\sim \times 2$ training and inference compute
- Slightly better performance

MODEL	#TOKENS	THROUGHPUT $\uparrow$	FID $\downarrow$
DiT-B/2	256	4.52	43.5
w/ ReDi (MR)	256	4.51	25.7
w/ ReDi (SP)	512	2.26	24.7

Main benchmark

Conditional Generation

- $\sim \times 6$ faster than REPA
- Converged performance :
 - REPA : 5.9 FID
 - ReDi : 3.3 FID

DiT-L/2	458M	400K	23.2
w/ REPA	458M	400K	15.6
w/ ReDi (ours)	458M	400K	10.5
SiT-L/2	458M	400K	18.5
w/ REPA	458M	400K	9.7
w/ ReDi (ours)	458M	400K	9.4
DiT-XL/2	675M	400K	19.5
w/ REPA	675M	400K	12.3
DiT-XL/2	675M	7M	9.6
w/ REPA	675M	850K	9.6
w/ ReDi (ours)	675M	400K	8.7
SiT-XL/2	675M	400K	17.2
w/ REPA	675M	400K	7.9
w/ ReDi (ours)	675M	400K	7.5
SiT-XL/2	675M	7M	8.3
w/ REPA	675M	4M	5.9
w/ ReDi (ours)	675M	700K	5.6
w/ ReDi (ours)	675M	4M	3.3

Main benchmark

Conditional Generation w/ CFG

MODEL	EPOCHS	FID↓	sFID↓	IS↑	PRE.↑	REC.↑
<i>Autoregressive Models</i>						
VAR	350	1.80	-	365.4	0.83	0.57
MagViTv2	1080	1.78	-	319.4	0.83	0.57
MAR	800	1.55	-	303.7	0.81	0.62
<i>Latent Diffusion Models</i>						
LDM	200	3.60	-	247.7	0.87	0.48
U-ViT-H/2	240	2.29	5.68	263.9	0.82	0.57
DiT-XL/2	1400	2.27	4.60	278.2	0.83	0.57
MaskDiT	1600	2.28	5.67	276.6	0.80	0.61
SD-DiT	480	3.23	-	-	-	-
SiT-XL/2	1400	2.06	4.50	270.3	0.82	0.59
FasterDiT	400	2.03	4.63	264.0	0.81	0.60
MDT	1300	1.79	4.57	283.0	0.81	0.61
<i>Leveraging Visual Representations</i>						
REPA	800	1.80	4.50	284.0	0.81	0.61
ReDi (ours)	350	1.72	4.68	278.7	0.77	0.63
ReDi (ours)	800	1.61	4.66	295.1	0.78	0.64

Selected Samples

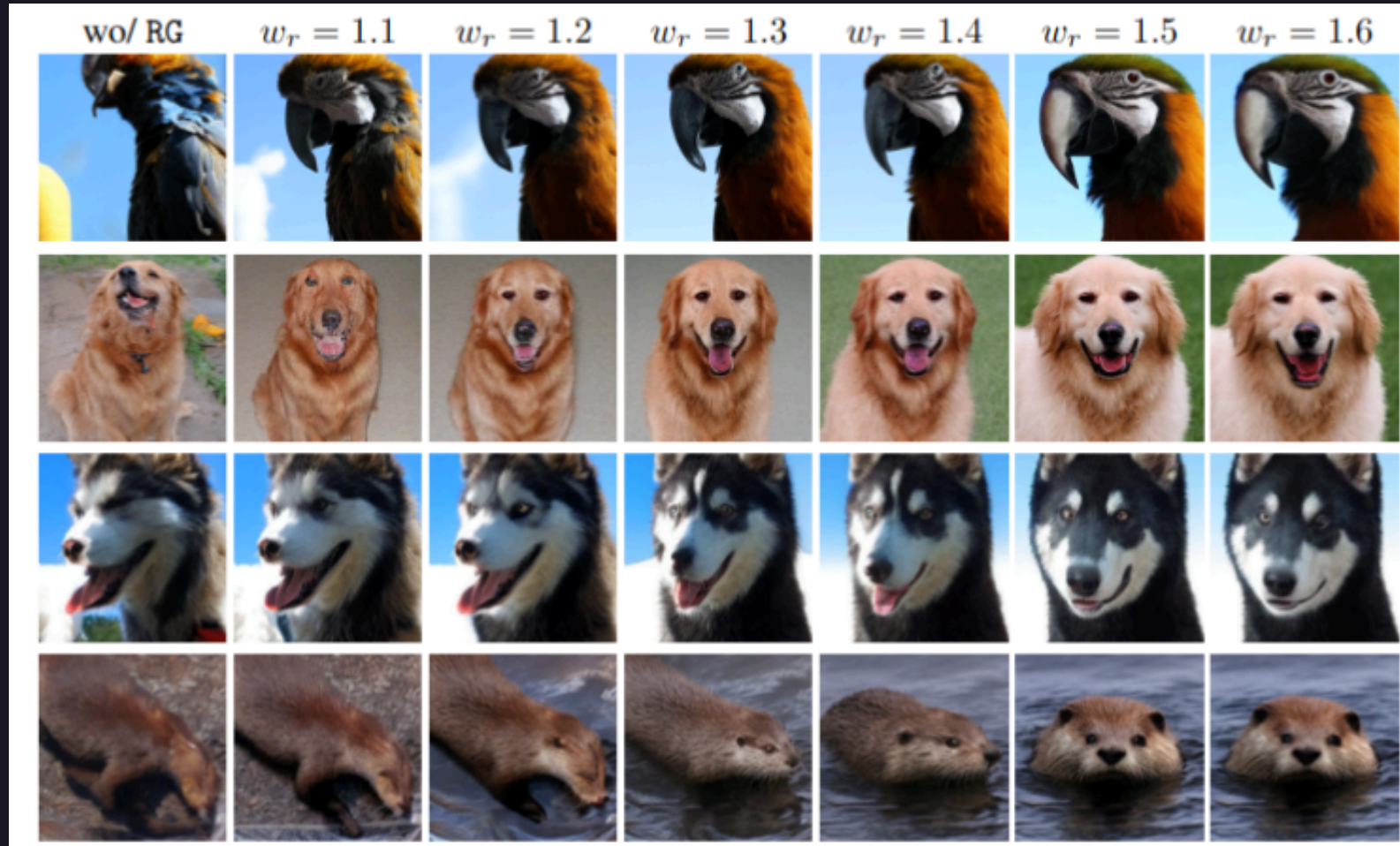


Unconditional Generation

- Representation Guidance (RG) is very usefull for unconditional generation
- ReDi significantly closes the gap between conditional and unconditional generation

MODEL	#PARAMS	FID↓
DiT-B/2 (conditional)	130M	43.5
DiT-B/2	130M	69.3
w/ ReDi (ours)	130M	51.7
w/ ReDi+RG (ours)	130M	47.3
DiT-XL/2 (conditional)	675M	19.5
DiT-XL/2	675M	44.6
w/ ReDi (ours)	675M	25.1
w/ ReDi+RG (ours)	675M	22.6

Representation Guidance



Possible Future Research Directions

- Multiple Representations (e.g. DINOv2 and CLIP)
- Better compression approach for DINOv2
- Leverage the generated representations

Thank You 😊